Title: Localized Embedding Models for Latency Reduction in RAG Architectures

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Abstract Retrieval-Augmented Generation (RAG) systems frequently suffer from network latency when relying on cloud-based APIs for text embeddings. This brief study explores the performance benefits of migrating from cloud services to localized models. By utilizing local models, systems can achieve near-instantaneous document ingestion and significantly faster query resolution.

1. Introduction In modern full-stack web applications, processing large PDF documents requires significant computational overhead. When a user uploads a dataset, the text must be extracted, chunked, and converted into high-dimensional vector arrays. Sending these chunks to external cloud APIs introduces network bottlenecks.

2. Methodology To eliminate this latency, we propose a localized architecture. Instead of transmitting data over HTTP to remote servers, the system routes text chunks to a locally hosted embedding model, specifically `nomic-embed-text`. These embeddings are then stored directly into a NoSQL database utilizing a vector search index.

3. Results and Conclusion By shifting the embedding process to the edge, the application completely bypasses external network latency during the ingestion phase. Furthermore, querying the database and generating summaries using localized models, such as `llama3`, provides a seamless, real-time user experience. This architecture proves highly effective for secure, high-speed academic research platforms.